

Section 7.4: Control Applications (Turtle Graphics)

Computers can be given **sets of instructions** which they will follow to **control objects on the screen**. Giving instructions to a computer is also known as '**programming**'.

In this section we will take a look at how a computer can be programmed using a simple language called '**LOGO**'.

Key Concepts of this section:

- # Understand how computers can be programmed to '**control**' objects on the screen.
- # Know the most common **LOGO commands**.
- # Be able to **use LOGO to draw different shapes** by inputting commands.

LOGO (aka Turtle Graphics)

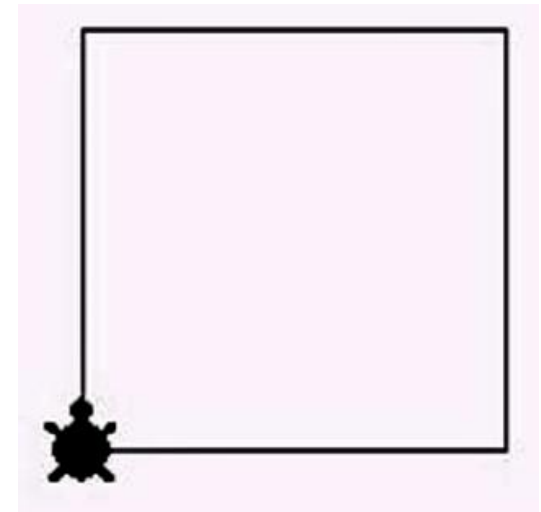
What is LOGO?

- # LOGO (**also known as 'Turtle Graphics'**) was developed in 1967 as a way to teach children **basic computer programming**.
- # In LOGO, a curser (**called a turtle**) can be controlled and moved around the screen by inputting simple commands. As the curser moves around the screen it draws a line.
- # Common LOGO commands include:



When logo was first released in the 1960's, commands controlled a **robot that was shaped like a turtle**.

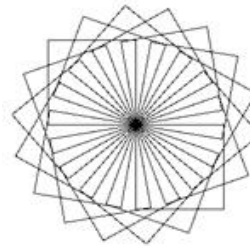
Command	Meaning	Example
PENDOWN	Lower the pen down (to draw a line)	PENDOWN
PENUP	Lift the pen up (to not draw a line)	PENUP
FORWARD n	Move the turtle forward n steps	FORWARD 50
BACKWARD n	Move the turtle backward n steps	BACKWARD 30
LEFT n	Turn left n degrees	LEFT 90
RIGHT n	Turn right n degrees	RIGHT 45
REPEAT n	Commands in between these lines should be repeated n times	REPEAT 4
ENDREPEAT		FORWARD 20 RIGHT 90 ENDREPEAT



Modern versions of LOGO have seen the physical robot turtle replaced with an **on-screen cursor turtle**

NOTE!

Using **combinations of these commands** makes it possible to draw all kinds of **different shapes**



Examples of LOGO in use

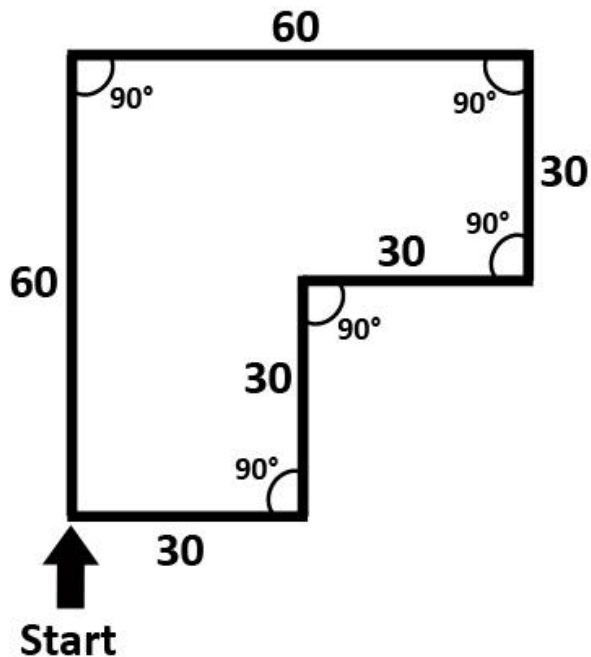
Here are some examples of LOGO commands being used to draw simple shapes. You will practice drawing each shape in an online version of LOGO which can be accessed by clicking the link in the grey box found on the right of each example.

Example one - a solid shape:

Shape to draw

LOGO commands to draw it

PENDOWN
FORWARD **60**
RIGHT **90**
FORWARD **60**



```
RIGHT 90
FORWARD 30
RIGHT 90
FORWARD 30
LEFT 90
FORWARD 30
RIGHT 90
FORWARD 30
```

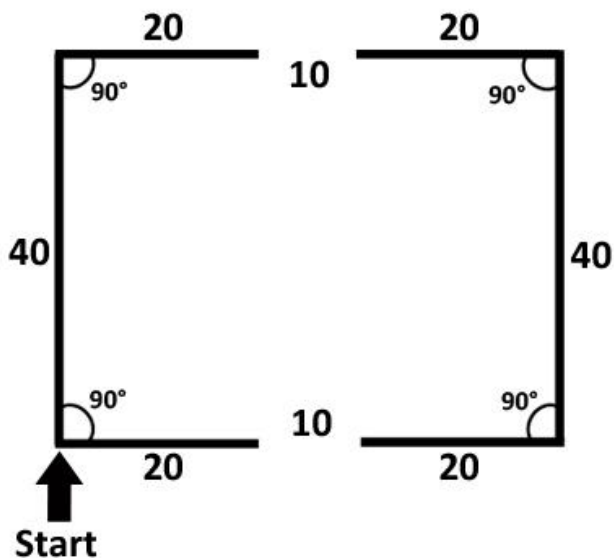
Click [here](#) to open an online version of LOGO and draw the shape to the left

NOTE:

In this version of logo the forward, right and left commands are abbreviated
(Forward = **FD**, Right = **RT**, Left = **LT**)

Example two - a shape with gaps:

Shape to draw



LOGO commands to draw it

```
PENDOWN
FORWARD 40
RIGHT 90
FORWARD 20
PENUP
FORWARD 10
PENDOWN
FORWARD 20
RIGHT 90
FORWARD 40
RIGHT 90
FORWARD 20
PENUP
FORWARD 10
PENDOWN
```

FORWARD 20

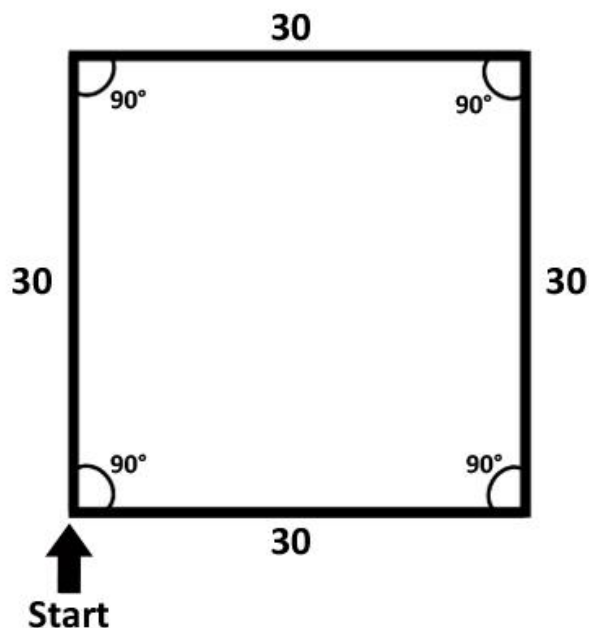
Click [here](#) to open an online version of LOGO and draw the shape to the left

NOTE:

In this version of logo the forward, right and left commands are abbreviated
(Forward = **FD**, Right = **RT**, Left = **LT**)

Example three - using REPEAT n ENDREPEAT:

Shape to draw



LOGO commands to draw it

Option 1

```
PENDOWN
FORWARD 30
RIGHT 90
FORWARD 30
RIGHT 90
FORWARD 30
RIGHT 90
FORWARD 30
```

Option 2

```
PENDOWN
REPEAT 4
FORWARD 30
RIGHT 90
ENDREPEAT
```

NOTE!

Both of the options above will draw the shape to the left. Option 2, however, does it in less steps by using the **REPEAT n ENDREPEAT COMMAND**

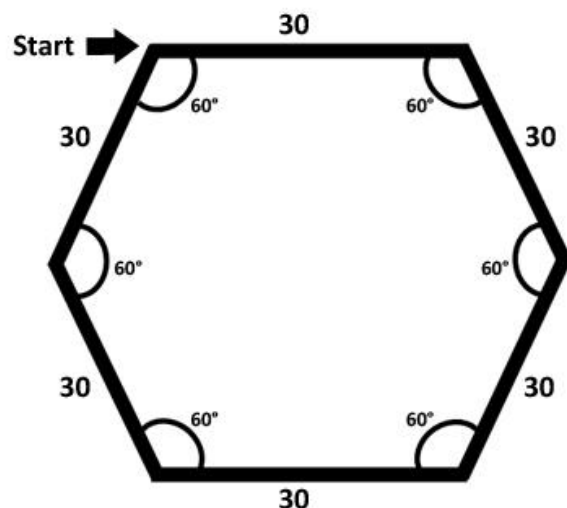
Click [here](#) to open an online version of LOGO and draw the shape to the left

NOTE:

In this version of logo the forward, right and left commands are abbreviated
(Forward = **FD**, Right = **RT**, Left = **LT**)

Example four - your practice shape:

Shape to draw



Click [here](#) to open an online version of LOGO and draw the shape to the left

Practice drawing it **both the long way** and using **REPEAT n ENDREPEAT**

Remember that in this version of LOGO some commands are entered slightly differently
(Forward = **FD**, Right = **RT**, Left = **LT**)

Activity!

[Past Paper LOGO Style Questions](#)

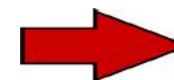
Click the above task and print all pages. Answer each of the past paper LOGO questions.

Answers to the questions can be found on pages 6 and 7. Check your answers and make any necessary changes.



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[Next - Control Devices Examples](#)



Links to Theory Units:

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[Section 7: The Ways in which ICT is used](#)

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[Section 7: Part 2](#)

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